

Liquidity Games in HFT Markets

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Outline

1. **TODAY**

- Project Introduction
- Initial Data Analysis (IDA)

2. **Tuesday & Wednesday**

- Exploratory Data Analysis (EDA)
- Model Development

3. **Thursday**

- Finite Size Scaling Analysis
- Questions

4. **Friday**

- Presentation: 30min Per Group

Objectives

1. Understand how methods from Statistical Physics are applied in empirical finance
2. Understand stylised facts of financial time series
3. Work with real high-frequency data: clean, filter, transform
4. Derive Microstructure features
5. Fit a model and infer a functional relationship

What is Market Microstructure?

The microstructure of a market is characterised by:

- Interactions amongst different kinds of participants,
- Rules governing exchanges, as defined by regulators.

These rules focus on

- Minimising any friction arising at the level of trading venues,
- Governing how the exchange of assets takes place in very specific settings.



FIG 1. Cover of LIFE issue of June 8, 1962, following the market crash of May 1962, which ran with the headline: "What went wrong in the wild stock market and what it means for the U.S.".

Limit Order Book

LOB

A LOB $L(t)$ is a dynamic system of active orders in a market at time t , where orders reside in queues at different price levels

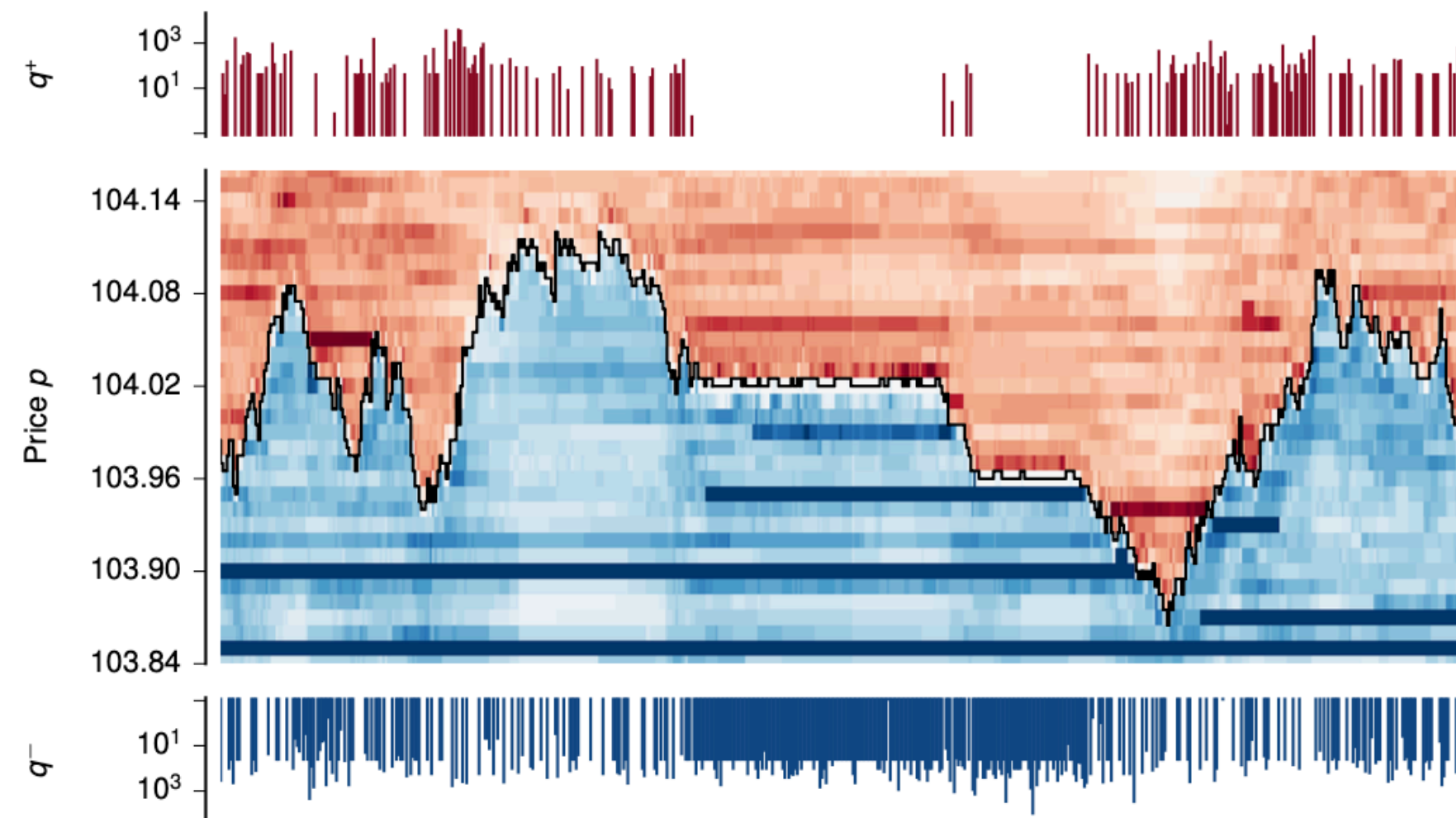


FIG 2. AAPL, May first 2016. The visible liquidity at each price level, just before each trade, is shown in red above and in blue below the mid-price (upper solid line). Full saturation corresponds to the 90th percentile for the day.

Bid-, Ask-Price

LOB MECHANISMS

At times, it is useful to consider $b(t)$ and $a(t)$ with respect to each other, where bid-price $b(t)$

$$b(t) := \arg \max_{x \in \mathcal{B}(t)} p_x$$

define the highest stated price among levels at each consecutive price, and the ask-price $a(t)$ at time t denoted:

$$a(t) := \arg \min_{x \in \mathcal{A}(t)} p_x.$$

Spread & Mid-price

At times, it is useful to consider $b(t)$ and $a(t)$ with respect to each other. Their difference:

$$s(t) := a(t) - b(t) > 0,$$

is the bid-ask spread (or simply the spread), which determines the cost (or, inversely, the profitability) of an instantaneous round trip (e.g., a buy immediately followed by a sell order).

Their mean:

$$m(t) := [a(t) + b(t)]/2,$$

is the mid-price at time t which we use as our choice of fair/reference price.

Orders

ORDERS:

An order x is a commitment to buy or sell a given volume of an asset. An order $x := (\varepsilon_x, p_x, v_x, t_x, \pi_x)$ is thus a tuple defined by five attributes:

- Its sign (or direction) $\varepsilon_x = \pm 1$,
- Price p_x ,
- Volume v_x ,
- Time of submission or execution t_x ,
- and order type π_x .

Order types

MO
LO
CA
QA

- MO is an arriving market order (trade execution).
- LO correspond to the arrival of a limit order.
- CA denotes (full or partial) cancellations of the length or volume of the queue at the best quotes.
- QA is an active order resting at some queue in the book.

In a LOB, an agent's intentions can be expressed using an assortment of different order types that represent Trades and Quotes:

Order Flow

In a LOB, liquidity provision (market-making) and taking are self-organising processes in which agents place orders on both the bid and ask sides of the queue.

- This complex inflow of limit orders (liquidity) and market orders (order flow) provides a tangible way to quantify assets' visible demand.
- Most traders assess the market depth at best quotes by looking at the volume profile of active orders $V_{b,a}$.

THE LENGTH OF BID SIDE INITIATED ORDERS:

$$\mathcal{E}_b(p, t) := \sum_{\{x \in \mathcal{B}(t) | p_x = p\}} | \varepsilon_x |.$$

VOLUME PROFILES:

$$V_b(p, t) := \sum_{\{x \in \mathcal{B}(t) | p_x = p\}} v_x.$$

Price Returns

In the investment world, there exist many different motives for investing.

- The most prominent is to earn a return on an investment:

$$\mathcal{R}(\ell) := (m_{t+\ell} - m_t) | \pi_t = \text{MO},$$

Price Returns

MO RETURNS:

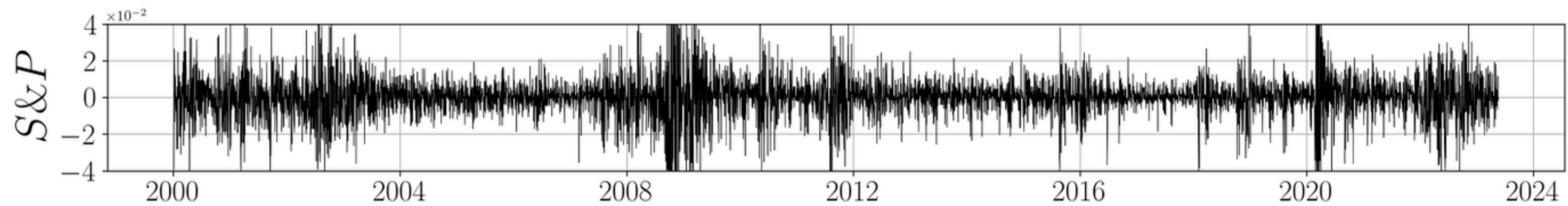


FIG 3. S&P500, Historic time-series of the S&P log-returns from 2000-2024, for trade executions (MO).

- Stationary Process
- Independent and Identically Distributed
- Exhibits peculiar empirical statistical properties

Returns – Stylized Facts

HEAVY TAILS & AGGREGATE GAUSSIANITY:

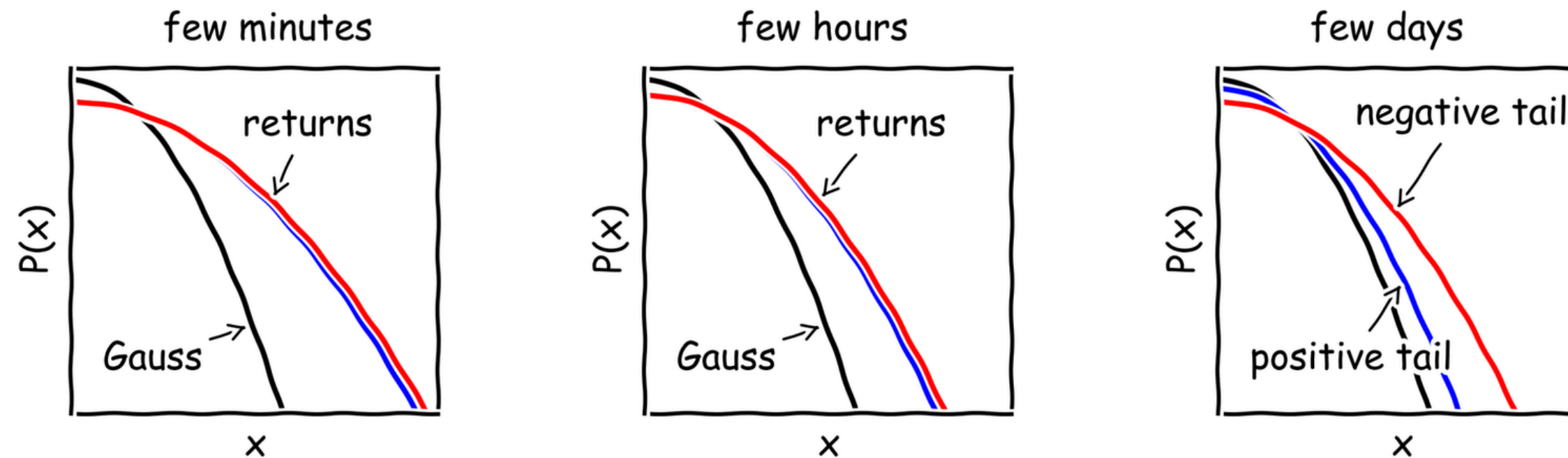
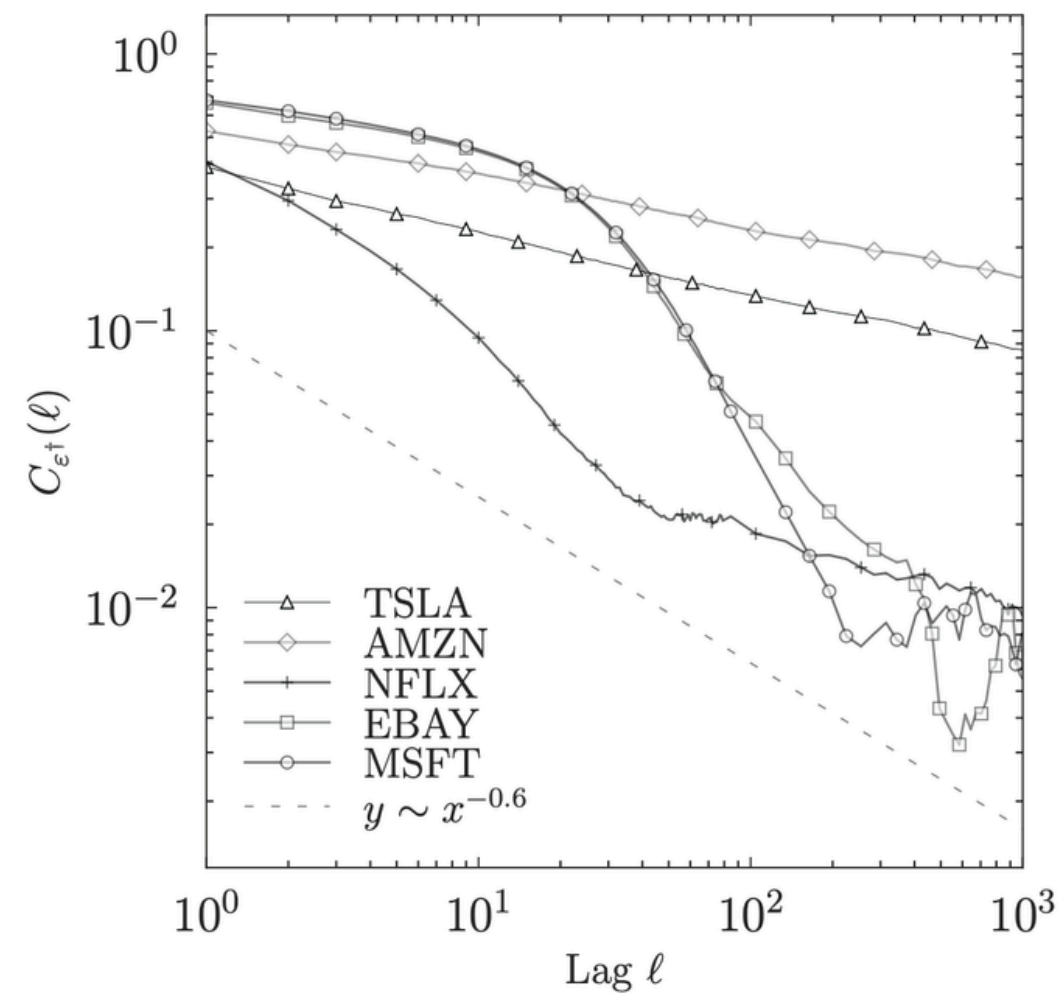


FIG 4. Distribution of returns on different timescales (log-scale).

- At high frequencies, extreme events that are many times bigger than the standard deviation occur at a much higher frequency than what would be expected if they were Gaussian distributed
- As one increases the time scale t over which returns are calculated, their distribution increasingly resembles a normal distribution. The distribution's shape differs across different time scales.

Returns – Stylized Facts



LONG RANGE CORRELATIONS

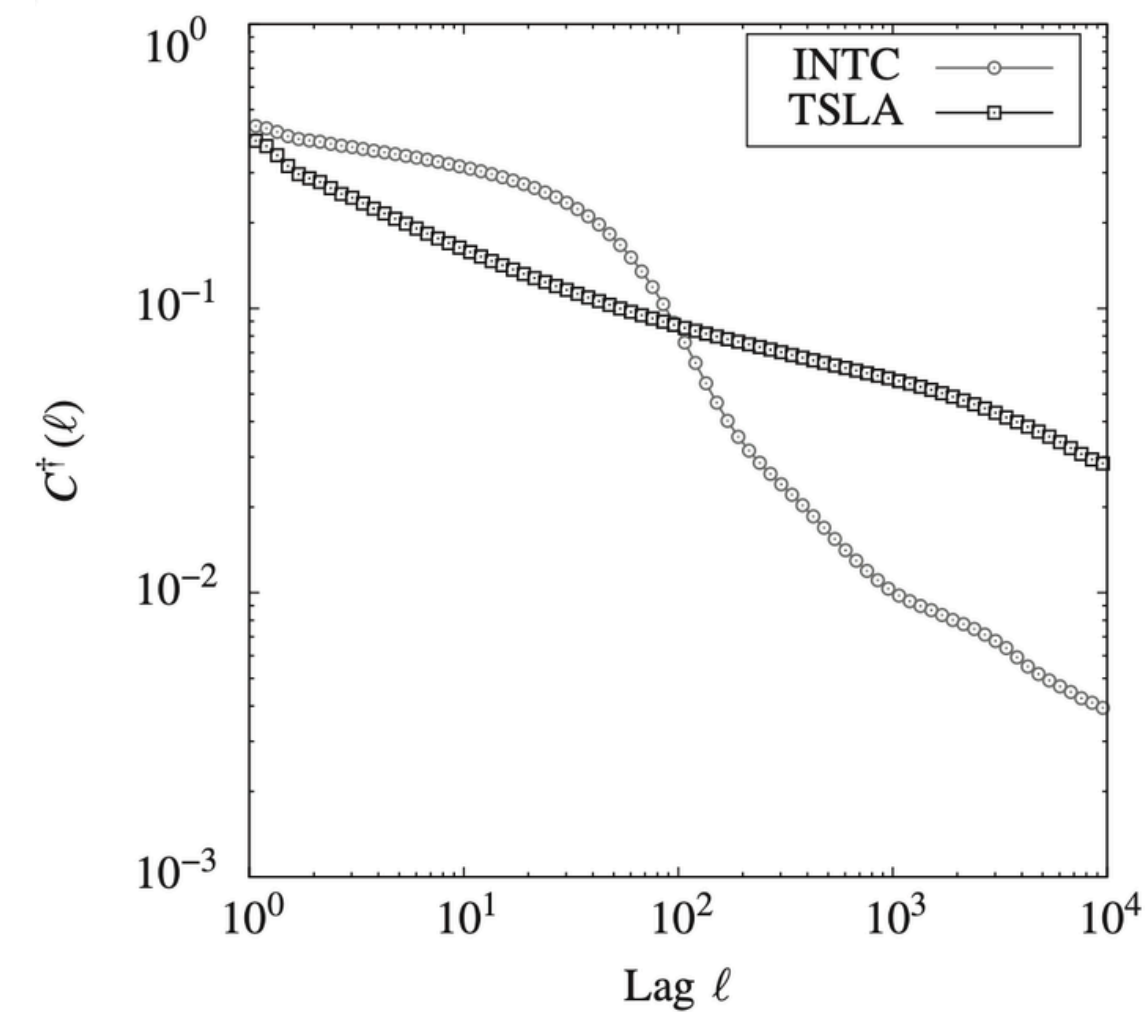


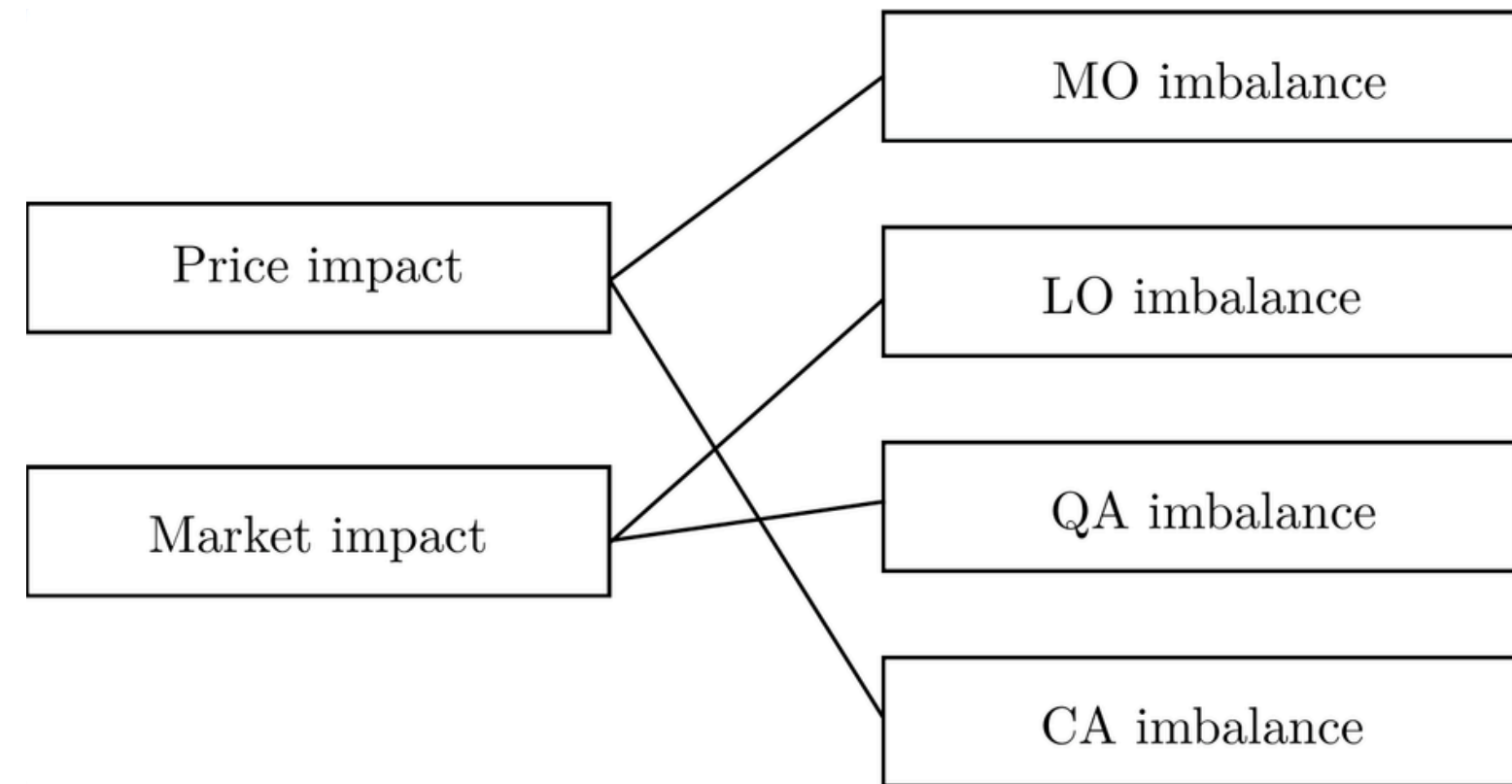
FIG 5. Autocorrelations of the log return (Right) magnitudes and signs (Left) for MO:

- That is, large price changes are typically followed by large ones and small changes by small ones.
- If we observe a buy (sell) market order now, the probability that a market order of 10,000 trades in the future is a buy (sell) is long-range predictable

Market Impact

Since order flow can be considered an expression of change in supply or demand,

- By studying market impact, we can more systematically quantify how the price changes with T.
- Conditional on the actions of different market participants through the use of impact functions.



IMPACT TAXONOMY

Unconditional Impact

Market impact is the correlation between an incoming order flow (to buy or to sell) and the subsequent price change:

$$\mathcal{R}_\pi(\ell) := \langle \varepsilon_t \cdot (m_{t+\ell} - m_t) | \pi_t = \text{MO} \rangle_t,$$

Recalling that m_t denotes the mid-price immediately before the arrival of the t th market order

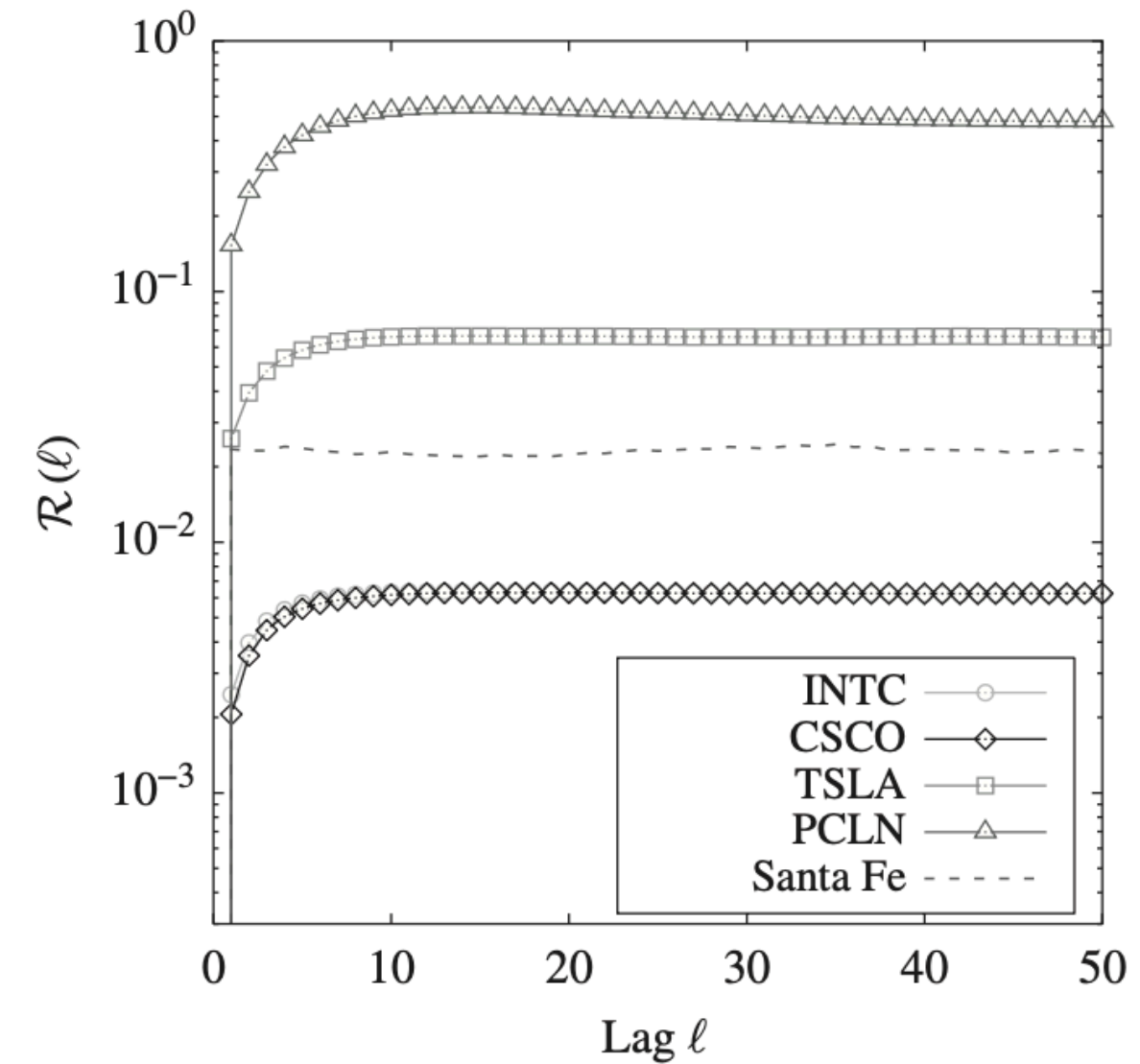


FIG 6. Response function $R(\ell)$ in market order event-time, for (circles) INTC, (diamonds) CSCO, (squares) TSLA, (triangles) PCLN and (dashed curve) in the simulated Santa Fe model.

Lag-1 Conditional Impact

So far, we have considered the impact of a market order irrespective of its volume. We define the lag-1 Conditional volume-impact as:

$$\mathcal{R}(v, 1) := \langle \varepsilon_t \cdot (m_{t+1} - m_t) | v_t = v \rangle_t,$$

It “seems” natural that large market orders should somehow impact prices more than small market orders.

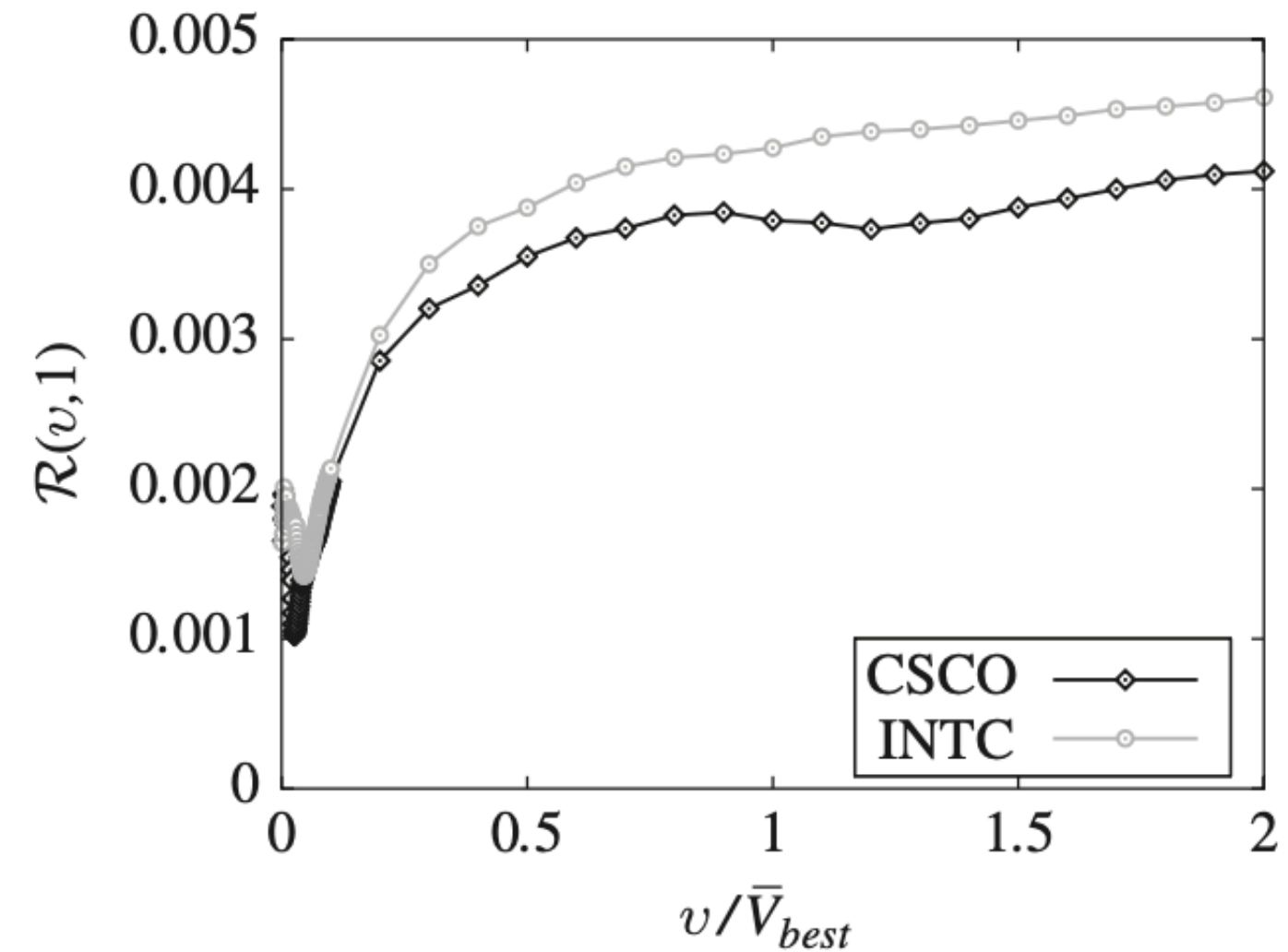


FIG 7. Lag-1 impact response function $\mathcal{R}(v,1)$ for INTC and CSCO, as a function of normalised market order volume (where we normalise by the average volume at the same-side queue).

Lag-1 Impact Conditional on Volume

However, after normalising, one finds that the volume-dependence of the immediate impact is actually strongly sub-linear:

$$\mathcal{R}(v, 1) := \langle \varepsilon_t \cdot (m_{t+1} - m_t) \mid v_t = v \rangle_t \cong A \left(\frac{v}{\bar{V}_{\text{best}}} \right)^\zeta \langle s \rangle_t.$$

where A is a constant of order 1 and ζ is an exponent that takes very small values.

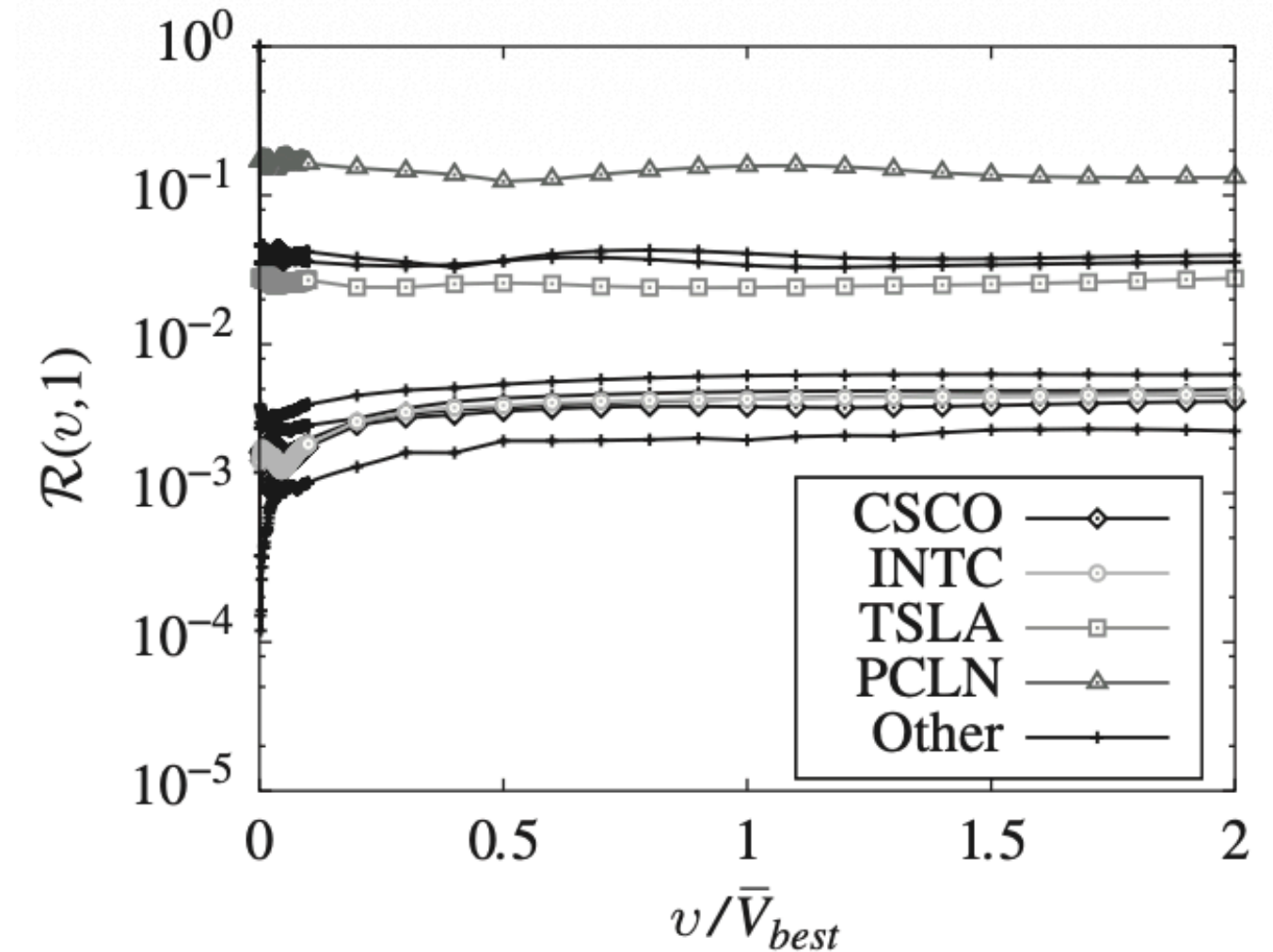


FIG 8. The same plot on a semi-logarithmic scale, for the ten stocks in Table 11.1. Small-tick stocks (with a large $\mathcal{R}(u, 1)$) do not exhibit any significant dependence on u

Aggregate Impact

Thus far, we have restricted our discussion exclusively to the dynamics of market returns at the most microscopic scale: individual buy and sell events

- To account for robust statistical regularities that can emerge over time, we also analyse events over some coarse-graining interval $[t, t + T)$.
- And, to reduce the role of microstructural idiosyncrasies, price impact is often measured not at the trade-by-trade level
- But instead, we consider how returns scale over some coarse-grained time scale T .

Aggregate Impact

The aggregate-volume impact is the price change (returns) over the interval $[t, t+T)$ conditioned to a certain volume imbalance:

$$R(\Delta V, T) := \langle m_{t+T} - m_t | \Delta V \rangle,$$

- where ΔV is the Volume Imbalance,
- and $\langle \dots \rangle$ denotes an empirical average over all time windows containing T successive trades, executed the **same day**.

Order Flow Imbalance

SIGNED ORDER FLOW IMBALANCE:

$$\Delta \mathcal{E} = \sum_{\{n \in [t, t+T) \mid \pi_t = \pi\}} \varepsilon_n,$$

SIGNED VOLUME IMBALANCE:

$$\Delta V = \sum_{\{n \in [t, t+T) \mid \pi_t = \pi\}} \varepsilon_n v_n,$$

Order Flow Imbalance

Together, both measures estimate the relative strength of buying and selling pressure across coarse-grained order book events of different types. For MO,

- If $\Delta V > 0$, more buyers arrive within the given interval,
 - and one expects prices to rise (resp. fall for LO and QA).
- If $\Delta V < 0$, then a net positive selling pressure pushes the price downwards
- A value $\Delta V \approx 0$ corresponds to a situation in which the buying and selling pressure is balanced,
 - which occurs when the number of events/volume considered is of approximately equal density.

Dimensionality Analysis

After aggregation, the return and signed volume curve looks exactly the same at any intraday scale of observation.

- Such qualities give rise to **dimensionless quantities** whose values are expected to occur without a characteristic size or length scale,
- But instead display (roughly) self-similar structures at all scales.
- We take them as signatures of potential scale-free structure in the dynamics of price returns.

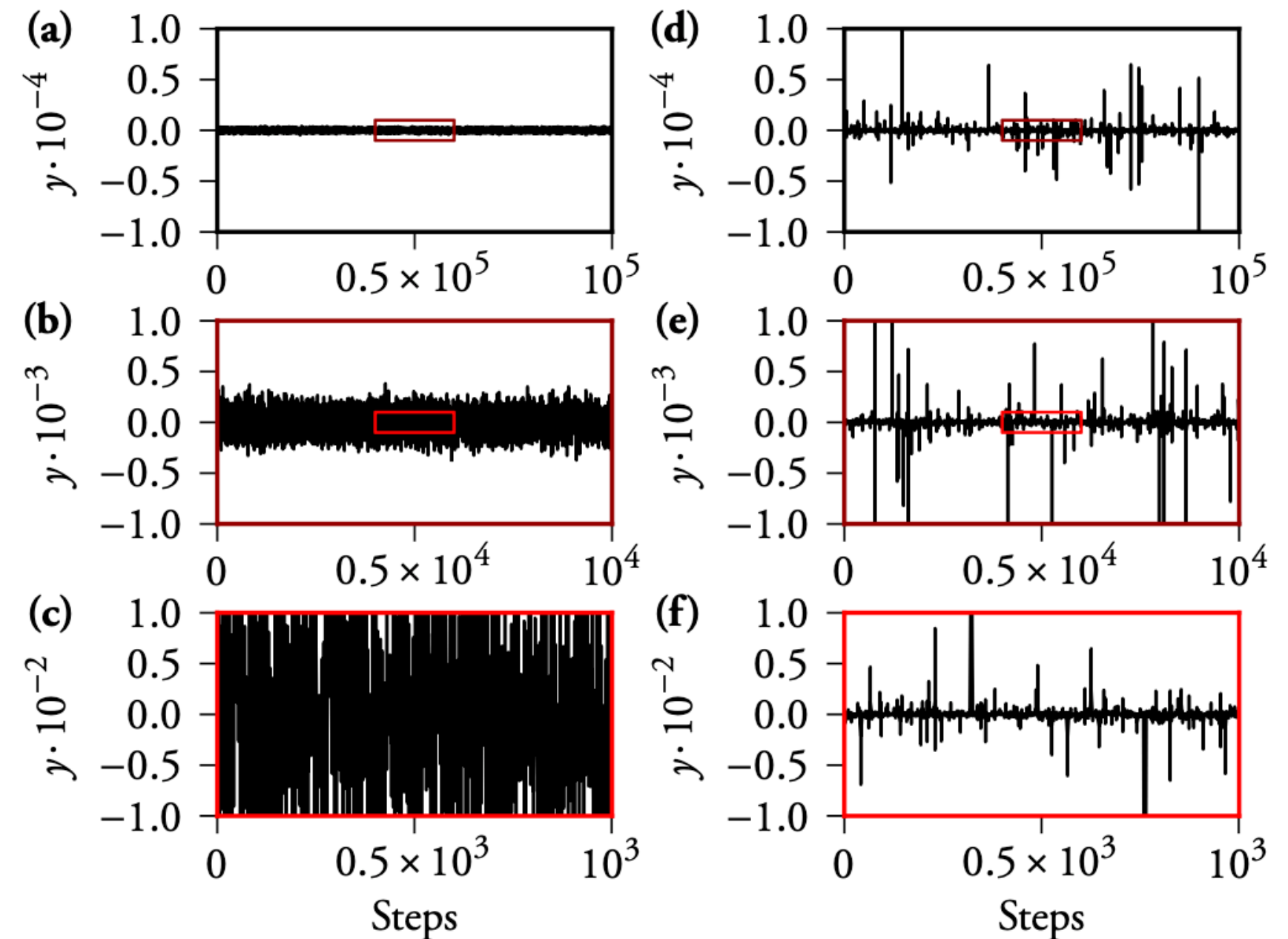
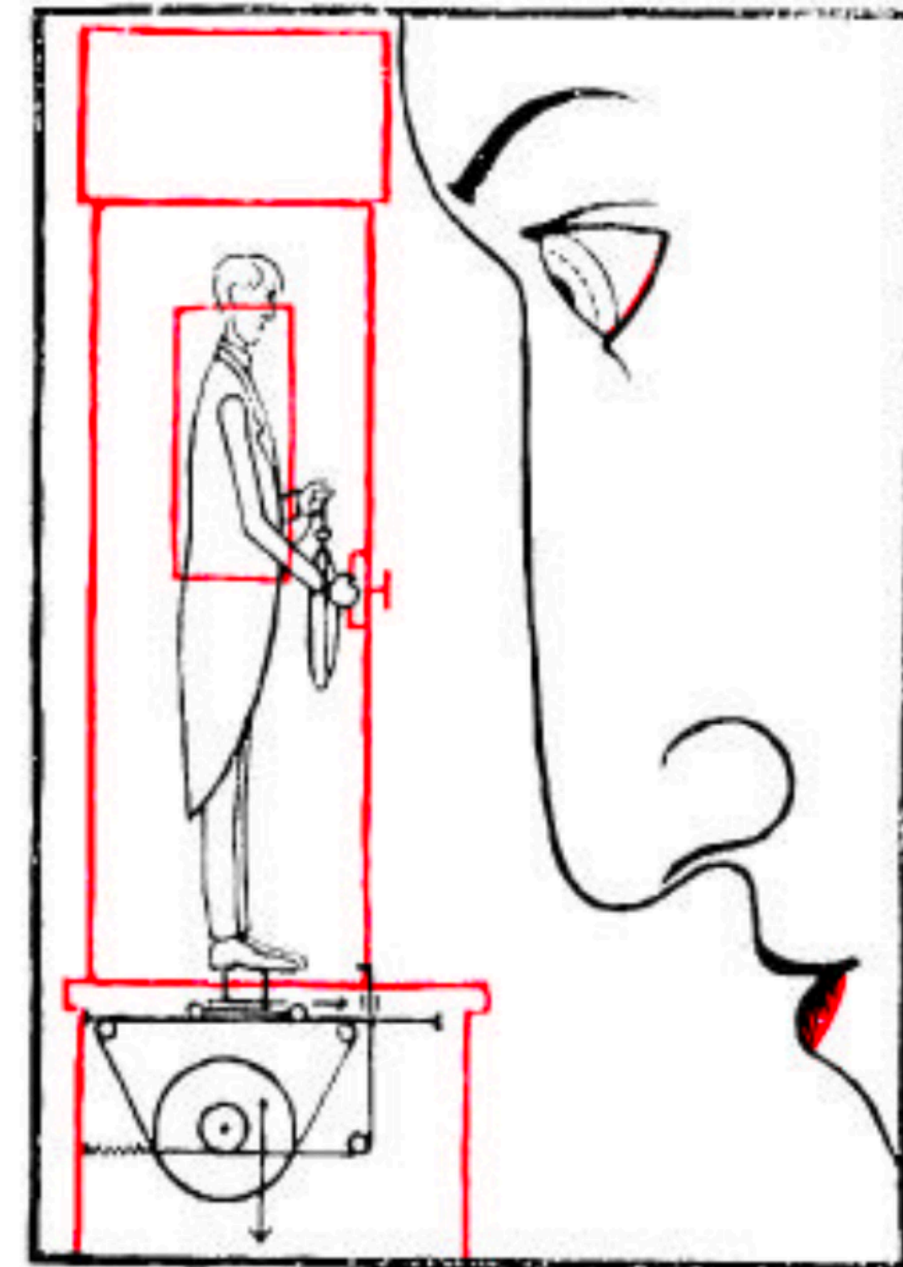


FIG 9. Two time-series at different magnitudes. **(a, b, c)**: Gaussian white noise. **(d, e, f)**: log-returns, which exhibit a power-law CCDF-tail with exponent $\alpha = [2-4]$ in the numeric limit for large fluctuations.

Dimensionality Analysis

Dimensionality analysis makes incomparable quantities comparable, even when originally expressed in different units.

- Even when dimensions are originally expressed at different scales
- Scaling laws can often be derived from dimensionality analysis through **Finite Size Scaling Analysis**
- This is supported by the **Renormalisation Group (RG)** argument/theory:
 - RG allows a systematic analysis of a system viewed at diff. scales
 - By marginalising from fine-grain measurement t to coarse-grained T .



Dimensionality Analysis

More formally, if a physical quantity y is to obey scaling (i.e., **is a power law**), it can be expressed as

$$A(x, T) = T^\psi \mathcal{F}(xT^{-\xi}),$$

- x is a variable describing a system
- T is the size of the system
- $\mathcal{F}(x)$ is a scaling function which too exhibits universal scaling
- ξ and ψ are dimensionless critical exponents that dictate the scaling of y and x , respectively.

Dimensionality Analysis

This gives a single effective theory (model) that describes the system at different length scales (i.e., a Master Equation):

$$A(x, T) = T^\psi \mathcal{F}(xT^{-\xi}),$$

- However, $F(x)$ cannot be determined by dimensional analysis
- Because we may not know the functional form of $F(x)$ on grounds of theory, statistical inference is acute.
- We must infer both the values of the critical exponents, as well as the form of the $F(x)$ itself, from data (**non-trivial**).

Dimensionality Analysis



FRACTALS ARE (APPROX). SCALE INVARIANT

Impact Scaling Laws

We are particularly interested in the scaling laws that govern the form of the conditional aggregate impact:

$$R(\Delta V, T) \cong R_T \cdot \mathcal{F}\left(\frac{\Delta V}{V_T}\right).$$

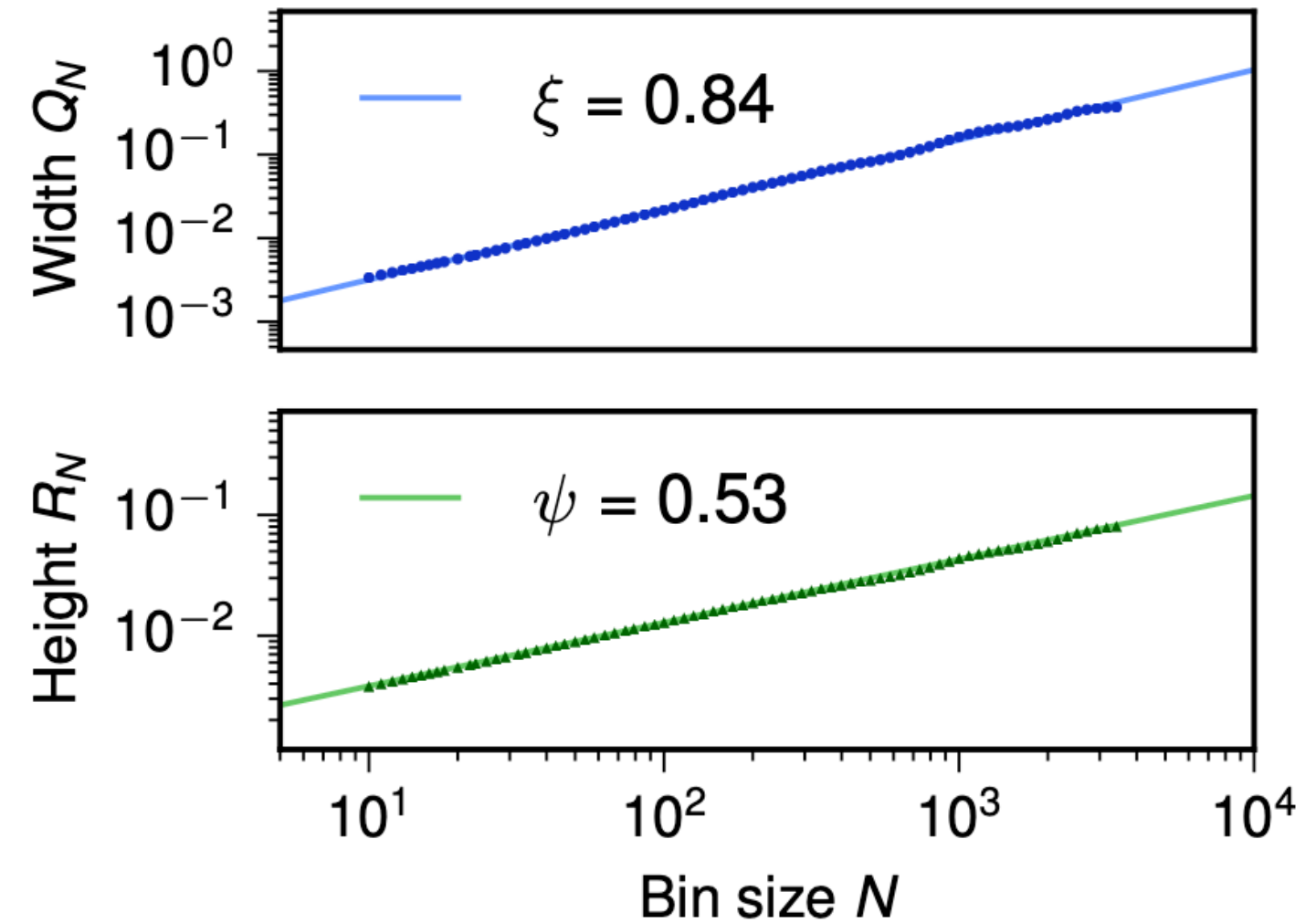


FIG 10. AAPL in 2016. **TOP** Volume scales VN for the impact curves. **Bottom**: The same analysis but for the return scale RN , i.e. the scaling of the impact curve along the return axis. Solid lines: power-law fits used for the rescaling along the volume-imbalance and returns axis.

Impact Scaling Laws

While width ΔV and height R of the impact curve scale independently with scaling given by scaling factors

$$\begin{aligned} V_T &\sim V_D T^\xi, \\ R_T &\sim \mathcal{R}(1) T^\psi, \end{aligned}$$

the following empirical **scaling law** holds for the Aggregate Volume Impact for a large variety of assets

$$R(\Delta V, T) \cong \mathcal{R}(1) T^\psi \mathcal{F}\left(\frac{\Delta V}{V_D T^\xi}\right).$$

where ξ and ψ are optimised scaling exponents, and V_D is the daily volume and $\mathcal{F}(\bullet)$ is a scaling function

Finite Size Scaling

Because different scales cannot be understood in isolation due to the data's correlation structure, we adopt the finite-size scaling method. Our objective is to:

1. Define the appropriate representation for the scaling function
2. Extract numeric values for the critical exponent from the data
 - a. There are several methods we can use to learn the model parameters from the data:
 - Nonlinear Least Squares (LS),
 - Maximum Likelihood Estimation (MLE),
 - Stochastic Gradient Descent (SGD).

Finite Size Scaling

Although $F(x)$ can in principle function of any form, Patzelt and Bouchaud (2017, 2018) and Farmer, Gerig and Lillo (2008) find it is well approximated by a sigmoidal:

$$\mathcal{F}(x) = \frac{x}{(1 + |x|^\alpha)^{\beta/\alpha}},$$

where α and β capture the mean and standard deviation that regulate the shape (steepness and symmetry) of $F(x)$.

Renormalisation

The height R and width ΔV are expressed in units of T by rescaling approach

$$\Delta V \longrightarrow \Delta V / V_T; \quad R \longrightarrow R / R_T.$$

After an appropriate averaging and renormalisation/transformation of $R(\Delta V, T)$, the self-similar data collapse onto a single Master Curve.

Renormalisation

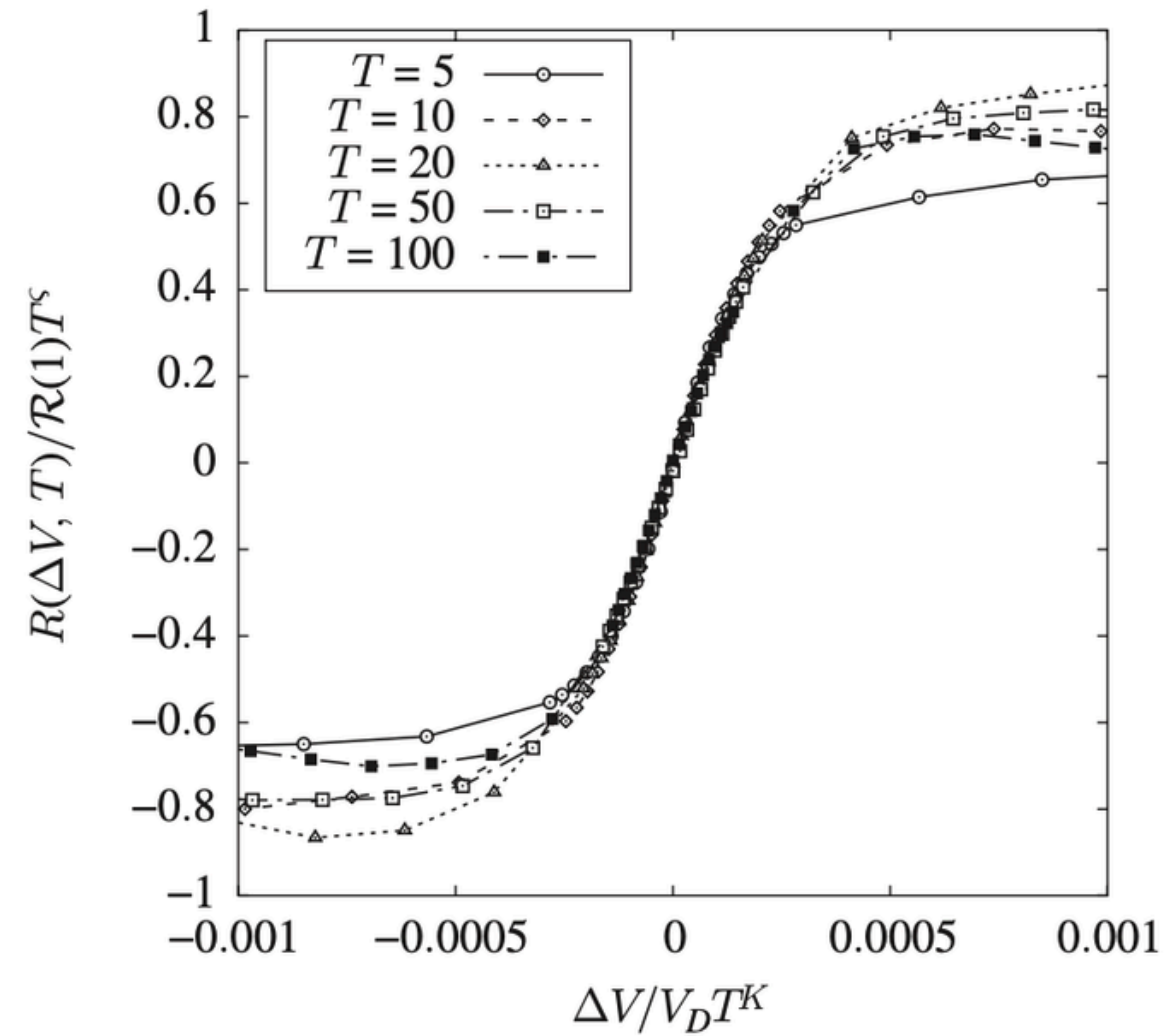


FIG 11. Aggregate impact scaling function $F(x)$ for TSLA. By rescaling $R(\Delta V, T)$ for $T=5, 10, 20, 50$ and 100 , the curves approx. collapse onto each other. Note that R and ΔV are rescaled each day by the corresponding values of $R(1)$ and the daily volume V_D .

Market Liquidity

Note that the slope of the linear region of aggregate impact curves follows;

$$\partial R(\Delta V, T) / \partial \Delta V = R_T / V_T.$$

It scales as $T^{-\lambda}$ i.e., it decreases with T as a power law T , with λ given by;

$$\lambda = \xi - \psi.$$

Market Liquidity

The slope of the linear region of $R(\Delta V, T)$ is usually called Kyle's lambda, in reference to the Kyle model.

- The value of this slope is often regarded as a measure of a market's (il-)liquidity

The larger the value of λ :

- The more a given imbalance in size/order flow affects the mid-price.
- The more expensive it is to trade.

Empirically, we see that when T increases, the dependence of $R(\Delta V, T)$ on the volume imbalance becomes closer to a linear relationship for small ΔV , while retaining its concavity for large $|\Delta V|$.

- The saturation for larger values of ΔV is also interesting, and is related to liquidity games:
 - Selective liquidity taking
 - Strategic liquidity provision

Questions?

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